

APEX TECHNOLOGIES

APEX PRO X1 Ultra

Official User Manual & Reference Guide

Model: APX-X1U-2025 | Series: Professional Elite

Intel Core Ultra 9	64 GB LPDDR5X	4 TB NVMe SSD
OLED 4K 120Hz	RTX 5090 Ti	Windows 11 Pro

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CHAPTER 1

Safety & Regulatory Information

Please read all safety instructions carefully before operating your APEX PRO X1 Ultra. Failure to comply with these guidelines may result in personal injury, property damage, or permanent damage to the device and will void the manufacturer warranty.

IMPORTANT WARNING

Never expose this device to water, rain, or high humidity.

Do not operate the laptop in environments where flammable gases are present.

Always place the laptop on a firm, flat, non-flammable surface during operation.

1.1 General Safety Guidelines

The APEX PRO X1 Ultra is a high-performance computing device designed for professional use. The following general safety guidelines must be observed at all times.

- Keep the device away from direct sunlight for extended periods, as prolonged heat exposure can permanently degrade battery capacity and display quality.
- Do not place heavy objects on top of the closed lid, as the display panel and internal components may suffer irreparable damage.
- Ensure adequate ventilation at all times. The bottom panel contains primary intake vents which must not be blocked during operation.
- Do not attempt to disassemble the device beyond what is explicitly permitted in the upgrade guidelines (Chapter 10). Unauthorized disassembly voids the warranty.
- If the device emits an unusual smell, smoke, or excessive heat, power it off immediately and contact APEX Support.
- This device is intended for use by individuals aged 18 years or older, or under adult supervision.

1.2 Electrical Safety

The APEX PRO X1 Ultra operates at a nominal input voltage of 20V DC via the proprietary APEX MagCharge 240W connector, or via USB-C Power Delivery (up to 140W). Electrical safety must be strictly observed.

- Use only the included APEX 240W GaN Power Adapter (Model: APX-PA240G). Using uncertified third-party adapters may cause fire, electric shock, or damage to the power circuitry.
- Inspect the power cable and adapter before each use. Do not operate the device if the cable is frayed, cut, or shows signs of physical damage.
- Do not connect the adapter to damaged or overloaded power outlets.
- Unplug the adapter during electrical storms to protect against power surges.
- The power adapter will generate heat during normal operation. Do not cover it during use.

1.3 Battery Safety

The APEX PRO X1 Ultra is equipped with an 8-cell, 99.9 Wh lithium-polymer battery. Lithium batteries store significant energy and require careful handling.

- Do not attempt to remove, puncture, crush, or incinerate the battery. The battery is a non-user-replaceable component sealed within the chassis.
- Avoid charging the device in ambient temperatures below 0 degrees Celsius or above 40 degrees Celsius.
- If the battery appears swollen, cracked, or deformed, immediately cease use and contact APEX Support for a battery replacement.
- Do not short-circuit the battery terminals under any circumstances.
- For long-term storage (more than 30 days), charge the battery to approximately 50 percent and store in a cool, dry environment.

1.4 Regulatory Compliance

- FCC Part 15 Class B (United States): This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.
- CE Marking (European Union): The APEX PRO X1 Ultra has been assessed and meets the requirements of applicable EU Directives including the Radio Equipment Directive (RED) 2014/53/EU and the RoHS Directive 2011/65/EU.
- BIS Certification (India): This product conforms to applicable BIS (Bureau of Indian Standards) standards and is approved for use in India.
- ENERGY STAR Qualified: This product has earned the ENERGY STAR certification for energy efficiency in laptop computers.

CHAPTER 2

Package Contents & Overview

Your APEX PRO X1 Ultra arrives in premium, ecologically responsible packaging. All packaging materials are recyclable. Please verify that all items listed below are present before disposing of the packaging.

2.1 In the Box

Item	Quantity
APEX PRO X1 Ultra Laptop	1 unit
APEX 240W GaN Power Adapter (APX-PA240G)	1 unit
Detachable 2.0m Power Cord	1 unit
USB-C to USB-C Charging Cable (2.0m, 240W rated)	1 unit
APEX Precision Cleaning Cloth (microfiber)	1 unit
Quick Start Guide	1 booklet
Warranty Card & Safety Sheet	1 document
APEX ProSleeve (premium neoprene sleeve)	1 unit

Note: If any item is missing or damaged, contact APEX Support within 7 days of purchase. Retain original packaging for warranty service.

2.2 Chassis Design

The APEX PRO X1 Ultra features a unibody CNC-machined aerospace-grade aluminium alloy (6061-T6) chassis with a Type III hard-anodized finish in Midnight Obsidian. The chassis has undergone MIL-STD-810H testing across 26 environmental conditions, ensuring durability under professional deployment scenarios.

Front & Lid

- Thin-bezel 16-inch OLED display occupying the entire lid surface, with a 93.2% screen-to-body ratio.
- Precision-etched APEX wordmark logo on the lid in APEX Signature Silver.
- Hidden camera module with a physical privacy shutter integrated into the top bezel.

Left Side

- APEX MagCharge 240W proprietary charging port.
- Thunderbolt 5 port x2 (40 Gbps, DP 2.1, Power Delivery).
- HDMI 2.1 port (up to 10K at 120Hz).
- 3.5mm Combo Audio Jack (headphone + microphone, 384 kHz / 32-bit).

Right Side

- USB-A 3.2 Gen 2 port x2 (10 Gbps).
- UHS-III SD Card reader (985 MB/s max).
- USB-C 4.0 port with DisplayPort 2.1 Alt Mode.

- Kensington Nano Security Lock slot.

2.3 Status Indicators

Power LED (White)	Solid white: device is powered on and active.
Power LED (Amber)	Solid amber: device is in Sleep mode.
Charge LED (Blue)	Pulsing blue: charging in progress.
Charge LED (Green)	Solid green: battery is fully charged (100%).
Charge LED (Amber)	Solid amber: battery below 20%, please connect charger.
Caps Lock Indicator	White LED embedded in the Caps Lock key.
Microphone Mute LED	Red LED on the F4 key indicating microphone is hardware-muted.
Camera Privacy LED	Amber LED adjacent to camera indicating physical shutter is closed.

CHAPTER 3

Getting Started

This chapter walks you through the initial setup of your APEX PRO X1 Ultra, from the first power-on sequence to configuring your operating environment for optimal performance.

3.1 First Boot Setup

Before powering on the device for the first time, connect the included 240W GaN adapter to ensure an uninterrupted setup experience. The device ships with approximately 40 percent battery charge for transport safety.

Step 1	Open the lid. The device will power on automatically due to the APEX AutoBoot feature. Alternatively, press the Power Button located in the top-right corner of the keyboard.
Step 2	The APEX splash screen will appear, followed by the Windows 11 OOBE (Out-of-Box Experience) setup wizard.
Step 3	Select your region, keyboard layout, and preferred language. These settings can be changed later in Windows Settings.
Step 4	Connect to a Wi-Fi network. This step is required to complete Windows Activation and to download initial driver updates.
Step 5	Sign in with your Microsoft Account or create a new one. For enterprise deployments, select 'Set up for work or school' and enter your organization credentials.
Step 6	Set up Windows Hello biometric authentication (face recognition or fingerprint). This step is strongly recommended for security.
Step 7	Choose your privacy settings for diagnostic data, location, and app permissions.
Step 8	Wait for the setup to finalize. The device may restart once or twice during this process.

3.2 Microsoft Account & Windows Hello

A Microsoft Account provides access to cloud-based services including OneDrive storage, Microsoft 365 apps (if subscribed), and synchronization of your personalized settings across devices. Windows Hello provides fast, secure, and passwordless authentication.

- Windows Hello Face Recognition uses the dual-camera IR array to authenticate in under 0.8 seconds, even in low-light conditions. This feature is exclusive to the APEX Pro X1 Ultra and cannot be spoofed by photographs.

- The integrated fingerprint reader (located in the power button) supports capacitive touch authentication. Register up to 10 fingerprints for multiple users or redundant finger registration.
- You may also set a PIN as a backup authentication method. APEX strongly recommends setting a minimum 8-character alphanumeric PIN.
- For enterprise environments, integration with Azure Active Directory (Azure AD) and single sign-on (SSO) is fully supported.

3.3 Driver & Firmware Updates

APEX provides regular driver and firmware updates through the APEX Update Manager, which is pre-installed on your system. Keeping your device updated ensures optimal stability, security, and performance.

- Open APEX Update Manager from the Start Menu or system tray icon upon first boot.
- Select 'Check for Updates' to scan for available driver, firmware, and BIOS updates.
- It is strongly recommended to install all Critical and Recommended updates before proceeding with daily use.
- BIOS updates must be performed with the device connected to AC power. Never interrupt a BIOS update, as this may render the device unbootable.
- Windows Update will run independently. Ensure both APEX Update Manager and Windows Update are completed before productive use.

CHAPTER 4

Display & Graphics

The APEX PRO X1 Ultra is equipped with a flagship-grade display and graphics subsystem, designed to deliver exceptional visual fidelity for creative professionals, engineers, and power users who demand color accuracy and visual performance in equal measure.

4.1 OLED Panel Technology

The 16-inch Samsung OLED panel used in the APEX PRO X1 Ultra represents the current pinnacle of laptop display technology, combining self-emissive pixel technology with professional color-science calibration.

Panel Type	OLED (Organic Light-Emitting Diode) — Self-emissive
Resolution	3840 x 2400 pixels (4K+ WQUXGA, 16:10 aspect ratio)
Pixel Density	283 PPI (Pixels Per Inch)
Refresh Rate	Up to 120 Hz (adaptive, via APEX VSync technology)
Response Time	0.1 ms (GTG — gray to gray)
Peak Brightness	1,000 nits (SDR), 1,600 nits (HDR Peak, Dolby Vision)
Typical Brightness	500 nits (SDR)
Contrast Ratio	2,000,000:1 (true infinite blacks via OLED)
Color Gamut	100% DCI-P3, 100% sRGB, 99% AdobeRGB
Color Depth	10-bit color (1.07 billion colors)
Delta E	< 1.0 (factory calibrated, X-Rite certified)
HDR Certification	DisplayHDR True Black 1600, Dolby Vision, HDR10+
Touch	No (deliberate design choice to minimize parallax for professionals)
Anti-Glare	Ultra-thin anti-reflective coating, < 0.5% reflectance

- OLED technology ensures each pixel generates its own light and can switch off completely, achieving true zero black levels that are physically impossible on LCD-based displays.
- The factory calibration process involves hardware-level LUT (Look-Up Table) profiling using a spectrophotometer, ensuring every unit ships with Delta E below 1.0 for professional color-critical work.
- A TUV Rheinland Low Blue Light certification ensures that extended viewing sessions are safer for eye health without compromising color accuracy.

4.2 GPU Architecture

The NVIDIA GeForce RTX 5090 Ti laptop GPU provides exceptional rasterization, ray tracing, and AI-accelerated compute performance, powered by the NVIDIA Blackwell architecture.

GPU Model	NVIDIA GeForce RTX 5090 Ti (Laptop)
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Architecture	NVIDIA Blackwell (5th generation)
CUDA Cores	10,496
RT Cores (4th Gen)	82 units
Tensor Cores (5th Gen)	328 units
VRAM	24 GB GDDR7 (256-bit memory bus)
Memory Bandwidth	896 GB/s
TGP (Total Graphics Power)	Up to 175W (configurable via APEX Control Center)
Max Resolution (HDMI)	10K @ 120Hz via HDMI 2.1
Max Resolution (DP)	16K @ 60Hz via DisplayPort 2.1
API Support	DirectX 12 Ultimate, Vulkan 1.3, OpenGL 4.6, OpenCL 3.0
DLSS Version	DLSS 4 (with Multi Frame Generation)
Encode / Decode	9th Gen NVENC, 9th Gen NVDEC (AV1, H.265, H.264)

4.3 Display Settings & Calibration

- Access display settings via APEX Control Center > Display > Color Profiles. Four factory-calibrated profiles are available: sRGB Standard, DCI-P3 Creative, AdobeRGB Professional, and Native OLED (unclamped).
- Refresh rate can be set to 60 Hz for battery efficiency or 120 Hz for maximum smoothness. APEX VSync Technology dynamically adjusts the refresh rate based on content type when set to 'Auto'.
- The 'OLED Care' feature is enabled by default and includes pixel shift, screen dimming on static content, and scheduled panel refresh to prevent image retention on long-duration static UI elements.
- Custom ICC profiles may be loaded via Windows Color Management for studio-calibrated workflows.

4.4 External Display Support

The APEX PRO X1 Ultra supports simultaneous connection to up to four external displays in addition to the built-in screen, using the available Thunderbolt 5, USB-C, and HDMI ports.

Max External Displays	4 simultaneously (plus built-in = 5 total)
HDMI 2.1 Output	Up to 10K @ 120Hz, or 4K @ 240Hz
Thunderbolt 5 (x2)	Each port: up to 8K @ 60Hz via DisplayPort 2.1 Alt Mode
USB-C 4.0	Up to 8K @ 30Hz or 4K @ 60Hz via DisplayPort 2.1
Daisy Chaining	Supported on Thunderbolt 5 ports (up to 2 displays per port)
G-SYNC Compatible	Yes (on external displays certified by NVIDIA)

CHAPTER 5

Keyboard, Touchpad & Input

The input system of the APEX PRO X1 Ultra has been engineered in collaboration with typing ergonomics researchers to deliver a keyboard experience that minimizes fatigue during extended coding or writing sessions, while the precision touchpad supports full Windows Precision Driver gestures.

5.1 Mechanical-Feel Keyboard

The keyboard features APEX TrueStrike technology — a low-profile scissor-switch mechanism with a membrane dampening layer, delivering the crisp tactile feedback of a mechanical keyboard with the slim profile required for a 14.9mm thin chassis.

Switch Type	APEX TrueStrike (Low-Profile Scissor with Dampening Layer)
Key Travel	1.5 mm actuation, 2.0 mm full travel
Actuation Force	45 gF (grams-force)
Layout	Full-size US QWERTY (100-key) with dedicated numpad row
Backlighting	Per-key RGB, 16.7 million colors, 5 brightness levels
Anti-Ghosting	Full N-Key Rollover (NKRO) — all keys simultaneous
Spill Resistance	IPX3 rated (up to 60mL, 15 seconds)

5.2 Per-Key RGB Lighting

- The per-key RGB backlighting is independently addressable using the APEX Aurora software, accessible from APEX Control Center.
- Twelve preset lighting effects are available out of the box, including Wave, Reactive, Breathing, Static Color, and Gaming Mode (highlights WASD cluster).
- Custom profiles can be saved and synchronized with game launchers or application contexts, so the keyboard lighting changes automatically when you switch between applications.
- Keyboard brightness is automatically reduced when running on battery to conserve energy. This behavior can be adjusted in APEX Control Center > Power > Input Lighting.

5.3 Precision Touchpad

Size	145 mm x 90 mm (class-leading for 16-inch laptops)
Technology	Capacitive Multi-Touch, Windows Precision Touchpad Driver
Surface Material	Etched Glass with oleophobic coating
Click Mechanism	Haptic feedback (forcepad), no physical moving parts
Multi-Touch Support	Up to 10 simultaneous touch points

Max Supported Gestures	25 (Windows standard + APEX custom gestures)
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Supported gestures include (but are not limited to):

- 2-finger scroll: Scroll in any direction on the active window.
- 2-finger pinch/zoom: Zoom in or out in supported applications.
- 3-finger swipe up: Open Task View (all open windows).
- 3-finger swipe down: Show Desktop.
- 3-finger swipe left/right: Switch between open applications.
- 4-finger tap: Open Action Center notification panel.
- 4-finger swipe left/right: Switch between virtual desktops.

5.4 Shortcut & Function Keys

F1	Mute / Unmute system audio
F2 / F3	Decrease / Increase speaker volume
F4	Hardware microphone mute (with status LED)
F5	Refresh / Reload in browser
F6	Toggle camera privacy shutter
F7 / F8 / F9	Media: Previous / Play-Pause / Next Track
F10	Toggle display output (Laptop only / Mirror / Extend)
F11	Decrease display brightness
F12	Increase display brightness
Fn + F1-F12	Standard F-Key behavior (configurable in BIOS)
Fn + Spacebar	Cycle keyboard backlight brightness levels
Fn + Q	Switch between APEX Performance Modes
Fn + Del	Print Screen (screenshot to clipboard)

CHAPTER 6

Ports & Connectivity

The APEX PRO X1 Ultra offers a comprehensive array of modern, high-bandwidth ports designed to eliminate the need for docking stations in most professional workflows, while also supporting enterprise docking via Thunderbolt 5.

6.1 Port Layout Summary

Left Side

- APEX MagCharge 240W (Proprietary Magnetic Charging)
- Thunderbolt 5 Port — Port 1 (40 Gbps, DP 2.1, PD 140W)
- Thunderbolt 5 Port — Port 2 (40 Gbps, DP 2.1, PD 140W)
- HDMI 2.1 Full-Size Port
- 3.5mm Combo Audio Jack (Hi-Res Audio, 384 kHz / 32-bit)

Right Side

- USB-A 3.2 Gen 2 (10 Gbps, 5V / 1.5A power output) — x2
- UHS-III SD Card Reader (985 MB/s read, 600 MB/s write)
- USB-C 4.0 with DisplayPort 2.1 Alt Mode
- Kensington Nano Security Lock Slot

6.2 Thunderbolt 5 & USB4

Both Thunderbolt 5 ports on the left side implement the Intel Thunderbolt 5 specification, which provides backward compatibility with Thunderbolt 4, Thunderbolt 3, USB4 Gen 3, USB 3.x, and USB 2.0 devices.

Data Bandwidth	40 Gbps bidirectional (scalable to 120 Gbps TX burst with Bandwidth Boost)
Power Delivery (Input)	Up to 140W (device charging from compatible PD charger)
Power Delivery (Output)	Up to 15W (charging connected peripherals)
Display Output	Up to two 8K @ 60Hz displays per port via DP 2.1
eGPU Support	Yes (via Thunderbolt 5 eGPU enclosures)
Daisy Chaining	Up to 6 Thunderbolt devices per chain
Connector Type	USB Type-C (reversible)

6.3 HDMI 2.1 & DisplayPort

- The full-size HDMI 2.1 port supports HDMI Forum VRR (Variable Refresh Rate), Quick Frame Transport (QFT), and Quick Media Switching (QMS) for use with gaming monitors and professional displays.
- Maximum output resolution via HDMI 2.1 is 10K at 120Hz or 4K at 240Hz for gaming use.

- DisplayPort 2.1 is available via the Thunderbolt 5 and USB-C 4.0 ports in Alt Mode. This standard provides 80 Gbps bandwidth, sufficient for a single 16K display or multiple 4K displays simultaneously.

6.4 SD Card Reader & Audio Jack

The UHS-III SD Card Reader supports SD, SDHC, and SDXC cards. With a rated read speed of 985 MB/s, it is capable of reading from high-speed V90-rated SD cards used in professional cameras and video equipment without data-rate bottlenecks.

The 3.5mm combo audio jack supports both CTIA and OMTP pinout standards via auto-detection. It is powered by a dedicated ESS Sabre ES9219C DAC/amplifier capable of driving headphones up to 600 Ohm impedance, making it suitable for audiophile-grade headphones directly without an external DAC.

CHAPTER 7

Wireless & Networking

The APEX PRO X1 Ultra features next-generation wireless connectivity, incorporating Wi-Fi 7 and Bluetooth 5.4, ensuring maximum throughput and minimal latency for both local area network and peripheral connectivity needs.

7.1 Wi-Fi 7 (802.11be)

Standard	IEEE 802.11be (Wi-Fi 7)
Backward Compatibility	Wi-Fi 6E, Wi-Fi 6, Wi-Fi 5, Wi-Fi 4
Frequency Bands	2.4 GHz, 5 GHz, 6 GHz (Tri-band simultaneous)
Max PHY Rate	Up to 5.8 Gbps (combined tri-band)
MIMO Configuration	4x4 MU-MIMO, 320 MHz channel width on 6 GHz
MLO (Multi-Link Operation)	Supported (simultaneous multi-band communication)
WPA Support	WPA3-Personal, WPA3-Enterprise, WPA2-Personal/Enterprise
Chip Vendor	Intel Wi-Fi 7 BE200 (M.2 2230 slot)
Antenna	2T4R (2 Transmit, 4 Receive) smart antenna system

- Wi-Fi 7 Multi-Link Operation (MLO) allows the device to simultaneously transmit and receive across multiple frequency bands, significantly reducing latency and improving reliability in congested environments.
- The 6 GHz band (Wi-Fi 6E/7) provides access to clean spectrum largely free of interference from legacy devices, resulting in more consistent performance in enterprise environments.
- APEX ProShield RF shielding on the antenna assembly reduces interference from internal components (GPU, NVMe, power circuitry) for improved wireless range.

7.2 Bluetooth 5.4

Standard	Bluetooth 5.4 (part of Intel BE200 combo chip)
Range	Up to 100m (Class 1, line of sight)
Max Throughput	2 Mbps (LE 2M PHY)
LE Audio	Supported (LC3 codec, Auracast broadcast audio)
Simultaneous Connections	Up to 7 active paired devices
Profiles	HFP, A2DP, AVRCP, HID, PAN, GATT, and more

7.3 Network Configuration

- To configure Wi-Fi: navigate to Settings > Network & Internet > Wi-Fi, select your network, and enter credentials. Enterprise networks with EAP-TLS/PEAP authentication are supported natively.

- For static IP configuration, navigate to Settings > Network & Internet > Wi-Fi > Hardware Properties > IP Assignment > Edit.
- The APEX PRO X1 Ultra does not include a physical RJ-45 Ethernet port. Wired Gigabit Ethernet is available via optional accessories: the APEX TB5 Pro Dock or the APEX USB-C to 2.5G Ethernet Adapter.
- Network bridging, ICS (Internet Connection Sharing), and Hyper-V virtual switch configurations are supported in Windows 11 Pro.

CHAPTER 8

Audio System

The APEX PRO X1 Ultra audio system has been designed in partnership with Harman Kardon, combining a high-output speaker array with Dolby Atmos spatial audio processing to deliver a soundstage that far exceeds conventional laptop audio.

8.1 Speaker Configuration

Configuration	Quad-speaker array: 2x tweeters (top-firing) + 2x woofers (bottom-firing)
Total Output Power	8W (2x 2W tweeters + 2x 2W woofers)
Frequency Response	55 Hz to 22 kHz (-3 dB)
Technology	Harman Kardon tuned, Dolby Atmos enabled
Smart Amplifier	TI TAS5825M (thermal and excursion protection)
Max SPL	89 dB SPL at 0.5m

- The top-firing tweeter array is positioned along the hinge to project sound upward toward the display, creating a more natural listening axis compared to downward-firing speaker designs.
- The bottom-firing woofers use resonance chambers tuned to maximize bass output from the constrained chassis volume, assisted by the TI smart amplifier which prevents overdrive distortion while maximizing volume output.
- Dolby Atmos software processing creates a three-dimensional soundscape that simulates overhead audio channels even on a stereo speaker system, particularly effective for cinematic content.

8.2 Microphone Array

Configuration	Quad-microphone array (4 mics) with beamforming
Noise Cancellation	AI-based noise suppression (APEX ClearVoice AI)
Frequency Response	100 Hz to 8 kHz
SNR (Signal-to-Noise Ratio)	> 65 dB (A-weighted)
Privacy Mute	Hardware mute via F4 key with physical LED indicator
Certification	Microsoft Teams Certified, Zoom Certified

Note: APEX ClearVoice AI is a real-time neural network-based noise suppression engine that runs locally on the Intel NPU (Neural Processing Unit) to eliminate background noise without consuming CPU resources or requiring an internet connection.

8.3 Audio Software (Dolby Atmos)

- The Dolby Access application (pre-installed) provides access to the Dolby Atmos processing engine. Access Settings > Sound > Dolby Atmos to configure the audio profile.

- Five Dolby Atmos presets are available: Dynamic (auto-adjusts), Music, Movie, Game, and Voice (optimized for calls and podcasts).
- Dolby Atmos is active for both built-in speakers and headphones. When headphones are connected, head-related transfer function (HRTF) processing provides headphone-specific spatialization.
- Third-party audio software (e.g., Equalizer APO, DTS:X) may conflict with Dolby Atmos. Only one spatial audio engine should be active at a time.

CHAPTER 9

Camera & Biometrics

The biometric and camera systems of the APEX PRO X1 Ultra are tightly integrated to provide seamless authentication and high-quality video communication, both protected by a hardware-level privacy shutter.

9.1 IR + RGB Camera Module

RGB Camera	13 MP Sony IMX258 sensor, f/2.0 aperture
IR Camera	940nm infrared for Windows Hello (class 2, eye-safe)
Video Output	Up to 4K @ 30fps (RGB), 720p @ 60fps (IR)
HDR	Supported in 1080p mode
Auto-Focus	Phase-detection autofocus (PDAF)
Low-Light Performance	f/2.0 + pixel binning, usable down to 5 lux
Privacy Shutter	Physical sliding shutter blocks both RGB and IR sensors
Interface	USB 2.0 internal (UVC compliant, driver-free on Windows 11)

Note: The physical privacy shutter mechanically blocks the camera optical path. It is independent of software. The amber camera privacy LED illuminates when the shutter is in the closed position, providing a clear visual indicator.

9.2 Windows Hello Face Login

- Windows Hello Face utilizes the IR camera to create a three-dimensional facial depth map, making it significantly more secure than 2D photo-based recognition systems.
- Authentication is performed locally on the device using the Intel IPU6 (Image Processing Unit) and the integrated NPU. No biometric data is ever transmitted to the internet.
- Face recognition works in darkness, low-light, and with most eyeglasses. It does not work with the privacy shutter closed (by design).
- To enroll: Settings > Accounts > Sign-in options > Windows Hello Face > Set up. Multiple lighting conditions can be enrolled for improved reliability.
- Sign-in speed is under 0.8 seconds from lid open to authenticated desktop on the APEX PRO X1 Ultra due to APEX InstantOn technology.

9.3 Fingerprint Reader

- The capacitive fingerprint reader is integrated into the power button on the top-right corner of the keyboard deck.
- The sensor supports 360-degree fingerprint reading — the finger can be placed in any orientation.
- Up to 10 fingerprints can be registered: Settings > Accounts > Sign-in options > Windows Hello Fingerprint.
- The fingerprint sensor communicates with the Windows Hello framework via a secure FIDO2-compliant channel. Fingerprint templates are stored in the TPM (Trusted Platform Module) and are never accessible to the OS or applications.

CHAPTER 10

Storage & Memory

The APEX PRO X1 Ultra is configured with enterprise-grade storage and memory subsystems, chosen to maximize both sequential throughput and random I/O performance which directly impacts real-world application launch times and multitasking responsiveness.

10.1 NVMe SSD Configuration

Primary SSD	Samsung 990 Pro 4 TB (M.2 2280, PCIe 5.0 x4 NVMe 2.0)
Interface	PCIe 5.0 x4 (128 Gbps bandwidth)
Sequential Read	Up to 14,500 MB/s
Sequential Write	Up to 13,000 MB/s
Random Read (4K)	Up to 2,500,000 IOPS
Random Write (4K)	Up to 2,200,000 IOPS
DRAM Cache	Yes (on-board LPDDR4 cache on SSD controller)
Secondary Slot	1x M.2 2280 PCIe 5.0 x4 slot (empty, user-upgradeable)
Encryption	AES-256 self-encrypting drive (SED), TCG Opal 2.0
Endurance (TBW)	3,600 TBW (Terabytes Written)

10.2 LPDDR5X RAM

Capacity	64 GB (soldered, dual-channel)
Type	LPDDR5X (Low Power Double Data Rate 5X)
Speed	8533 MT/s (JEDEC specification)
Bandwidth (Total)	~136 GB/s (dual-channel)
ECC Support	Supported on Intel Core Ultra 9 platform (soft ECC)
Max Supported (Platform)	96 GB (single SODIMM slot available for expansion)
Expansion Slot	1x SODIMM slot (accepts DDR5-5600 up to 32 GB)

10.3 Upgrade Guidelines

UPGRADE NOTICE

Upgrading internal components is permitted only by authorized APEX service personnel or technically qualified users.

Opening the device voids the ingress protection (IPX3) rating.

Always use an anti-static wrist strap and work on a non-conductive surface.

- The secondary M.2 2280 slot accepts PCIe 5.0 x4 NVMe drives. PCIe 4.0 x4 NVMe drives are also compatible but will operate at PCIe 4.0 speeds.
- The single SODIMM slot accepts DDR5-5600 or DDR5-4800 modules up to 32 GB. Combined with the soldered 64 GB, the maximum system RAM is 96 GB.
- To access internal components: remove the 10 Phillips PH00 screws from the bottom panel (two are hidden under the rear rubber feet) and use a plastic spudger to release the panel clips.
- After upgrading the SSD, Windows may require re-activation if significant hardware changes are detected. Ensure you have your Microsoft Account credentials available.

CHAPTER 11

Battery & Power Management

The APEX PRO X1 Ultra features a large-capacity battery combined with intelligent power management software to deliver the best possible runtime while maintaining system performance. Proper battery care practices extend the long-term health of the battery.

11.1 Battery Specifications

Cell Configuration	8-cell lithium polymer (Li-Po)
Rated Capacity	99.9 Wh (9,900 mAh @ 10.8V nominal)
Typical Battery Life	Up to 18 hours (video playback, APEX Efficiency Mode)
Typical Battery Life	Up to 11 hours (mixed productivity workload)
Minimum Life (Full Load)	Approx. 2 hours (CPU+GPU at maximum sustained load)
Charge Time (0-100%)	Approx. 90 minutes (240W adapter, Rapid Charge mode)
Charge Time (0-80%)	Approx. 42 minutes (240W adapter, Rapid Charge mode)
USB-C PD Charging	Up to 140W (via Thunderbolt 5 ports)
APEX Battery Health Guard	Limits charge to 80% by default to extend cycle life

11.2 Charging Modes

Rapid Charge (Default)

Uses the 240W adapter at full power. Reaches 80% in approximately 42 minutes. Generates the most heat and should be used when speed is the priority.

Balanced Charge

Charges at a reduced rate (100W equivalent) to minimize heat generation during long-duration stationary use with the adapter permanently connected.

APEX Battery Health Guard

Caps the maximum charge level at 80% of rated capacity. Significantly extends the total number of charge cycles (from ~500 to ~1000+ cycles at 80% threshold vs 100%).

Travel Mode Charge

Detects that charging is occurring via USB-C PD and adjusts power delivery to the maximum available from the source.

11.3 Power Plans & Battery Care

APEX provides three primary power plans accessible via APEX Control Center or the system tray battery icon. Each plan balances CPU/GPU TDP, screen brightness, peripheral power, and background activity differently.

Power Plan	CPU TDP	GPU Mode	Display	Battery Life
APEX Efficiency	15W (eco-boost)	iGPU only	60Hz, 400 nits	Up to 18h
APEX Balanced	28W adaptive	dGPU on-demand	Auto Hz, 500 nits	Up to 11h
APEX Performance	55W sustained	dGPU always-on	120Hz, 500 nits	Up to 6h
APEX Turbo	65W+ (burst)	dGPU max TGP 175W	120Hz, 1000 nits	Up to 2.5h

11.4 Hibernate & Sleep States

S0 (Active)	System fully operational. Normal power consumption.
S0ix (Modern Standby)	Ultra-low power idle while maintaining network connectivity. Resume in under 0.5 seconds.
S3 (Traditional Sleep)	RAM powered, CPU/GPU in deep sleep. Resume in 2-3 seconds. Can be enabled in BIOS for legacy compatibility.
S4 (Hibernate)	RAM contents saved to SSD, full power off. Resume from hibernate takes 8-12 seconds.
S5 (Soft Off)	Device is fully powered off. Cold boot takes approximately 6 seconds.

CHAPTER 12

Security Features

The APEX PRO X1 Ultra incorporates a layered hardware and software security architecture, designed to protect sensitive data and ensure platform integrity in enterprise and professional environments.

12.1 TPM 2.0 & Secure Boot

- The device is equipped with a dedicated Infineon SLB 9672 TPM 2.0 (Trusted Platform Module) chip. The TPM provides hardware-based cryptographic functions including key generation, secure key storage, and attestation.
- Secure Boot is enabled by default in the UEFI firmware. It verifies that only software cryptographically signed by trusted authorities (Microsoft and APEX Technologies) executes during the boot sequence, preventing rootkits and bootloader malware.
- The UEFI firmware supports Microsoft Pluton Security Processor integration, which moves the root of trust into the CPU package for even stronger isolation.
- To configure Secure Boot or TPM settings, access the APEX BIOS Setup Utility (press Delete or F2 during POST).

12.2 BitLocker Encryption

- Windows 11 Pro (pre-installed) includes BitLocker Drive Encryption, which can encrypt the entire system SSD using AES-256-XTS. When combined with the SED (Self-Encrypting Drive) capability of the Samsung 990 Pro, hardware-accelerated full-disk encryption is supported.
- BitLocker can be configured to use the TPM 2.0 chip as its key protector, meaning the drive will only decrypt when the correct trusted boot environment is detected.
- Optionally, a BitLocker PIN or USB startup key can be required in addition to TPM validation for a stronger two-factor pre-boot authentication model.
- BitLocker recovery keys should be backed up to your Microsoft Account, Active Directory, or a separately stored USB flash drive.

12.3 Hardware Privacy Shutter

A physical sliding shutter is built into the display bezel above the camera module. Sliding it to the closed position (indicated by the amber LED) physically blocks the optical path to both the RGB and IR camera sensors with an opaque mechanical barrier. This provides an absolute hardware guarantee that no software — including malware or OS vulnerabilities — can activate the camera when the shutter is closed.

12.4 Kensington Lock Slot

A Kensington Nano Security Lock slot is located on the right side panel. It is compatible with Kensington Nano locks and MicroSaver 2.0 locks. This provides physical security against opportunistic theft in shared workspaces, offices, or public environments.

CHAPTER 13

Performance & Thermal Management

The APEX PRO X1 Ultra uses an advanced multi-mode thermal architecture to manage the combined TDP of up to 240W (CPU + GPU combined) within a chassis that measures only 14.9mm at its thinnest point.

13.1 CPU & GPU Boost Profiles

CPU Model	Intel Core Ultra 9 285HX (24 cores: 8P + 16E + 4LP-E, up to 5.6 GHz)
CPU TDP Range	45W base, configurable 15W to 65W via APEX Control Center
CPU Boost (Short)	120W PL2 (Power Level 2) for up to 56 seconds
GPU TDP	Configurable 25W to 175W (Turbo mode)
Combined TDP Cap	240W total system (CPU + GPU simultaneous sustained)
NPU	Intel AI Boost (48 TOPS) — dedicated neural processing unit in CPU

13.2 Cooling Architecture

The thermal design uses a five-heatpipe vapor chamber architecture with dual 72-blade fans (fan diameter: 75mm each), exhausting via three vents (two rear, one right side). Liquid metal thermal compound is applied between the CPU die and the vapor chamber for maximum heat transfer efficiency.

- Vapor Chamber Cooling: A large-area vapor chamber covers both the CPU and GPU dies, spreading heat across the entire heatsink surface area rather than relying on point-to-point heatpipes alone.
- Dual 72-Blade Fans: High blade count and aerodynamic blade geometry reduce audible tonal noise while maintaining airflow volume. Fan RPM is independently controlled per-fan.
- Liquid Metal TIM: Applied by APEX during manufacturing, liquid metal thermal interface material (Thermal Conductivity > 73 W/mK) provides significantly better heat transfer than conventional thermal paste (4-8 W/mK typical).
- APEX Therma-Boost AI: A machine learning algorithm running on the Intel NPU predicts thermal load spikes and pre-emptively increases fan speed before thermal throttling can occur.
- Maximum fan speed is 6,500 RPM on CPU fan and 6,200 RPM on GPU fan, producing approximately 43 dB(A) at maximum RPM. Idle fan noise is inaudible at 0-200 RPM.

13.3 APEX Control Center

APEX Control Center is the unified performance and system management application. It provides real-time monitoring, profile switching, and hardware tuning in a single interface.

- Dashboard View: Real-time graphs for CPU temperature, GPU temperature, CPU utilization, GPU utilization, RAM usage, SSD temperature, fan speeds, and battery state.

- Performance Mode Switcher: Instantly switch between Efficiency, Balanced, Performance, and Turbo modes. Can also be triggered via the Fn+Q keyboard shortcut.
- Fan Control: Manual fan speed override (fixed RPM or percentage) for users who prefer predictable cooling over adaptive control.
- GPU Mode Selector: Switch between Auto (discrete GPU on demand), iGPU Only (best battery life), and Discrete Only (maximum performance) without requiring a restart.
- Keyboard Lighting: Full per-key RGB customization via APEX Aurora tab within Control Center.
- Power Delivery Monitor: Displays current charging rate (in Watts), battery state of health, estimated remaining cycles, and battery temperature.

CHAPTER 14

Software & Applications

The APEX PRO X1 Ultra ships with Windows 11 Pro (64-bit) and a curated set of pre-installed applications. APEX maintains a policy of minimal bloatware — only software with functional value for professional users is included.

14.1 Pre-installed Software

Application	Purpose	Developer
Windows 11 Pro (64-bit)	Operating System	Microsoft
APEX Control Center	Performance, Display, Fan & RGB Management	APEX Technologies
APEX Update Manager	Driver and Firmware Update Utility	APEX Technologies
APEX DiagTools	Hardware Diagnostic and Health Check Tool	APEX Technologies
Dolby Access	Dolby Atmos Audio Management	Dolby Laboratories
McAfee LiveSafe (30-day trial)	Antivirus & Security Suite	McAfee
Microsoft 365 (trial)	Office Applications Suite	Microsoft
Adobe Creative Cloud (trial)	Creative Professional Suite	Adobe
NVIDIA GeForce Experience	GPU Driver Updates and Game Optimization	NVIDIA
Intel Arc Control	Intel iGPU Management and Monitoring	Intel

14.2 APEX Control Center Deep Dive

APEX Control Center (ACC) is the primary interface between the user and the hardware capabilities of the APEX PRO X1 Ultra. Its modular panel structure provides quick access to all key subsystems.

- System tab: CPU, GPU, RAM, and SSD real-time telemetry. Includes a 60-second rolling graph and peak-value tracking per session.
- Display tab: Color profile selection, HDR toggle, refresh rate override, OLED Care settings, and external display detection.
- Power tab: Power plan selection, Battery Health Guard toggle, charging mode, and power delivery statistics.
- Thermal tab: Fan speed profiles, thermal threshold alerts, and Liquid Metal health indicator.
- Network tab: Wi-Fi band preference, Bluetooth device manager shortcut, and proxy configuration.
- Updates tab: Integrates with APEX Update Manager for one-click access to all available system updates.

14.3 Driver Management

- Always use APEX Update Manager as the primary update path for all hardware drivers. It ensures driver versions are tested against the APEX PRO X1 Ultra hardware combination.

- NVIDIA GeForce Experience can be used for GPU driver updates. However, only install NVIDIA Studio Drivers for professional workloads; Game Ready Drivers are optimized for gaming but may introduce instability in creative applications.
- Intel Driver & Support Assistant (IDSA) is available for Intel platform drivers but will default to APEX Update Manager recommendations when detected.
- Manual driver installation via Device Manager is supported but not recommended for non-technical users.

CHAPTER 15

Maintenance & Care

Proper maintenance of your APEX PRO X1 Ultra will preserve its aesthetic condition, thermal performance, and long-term reliability. The following guidelines are recommended by the APEX engineering team.

15.1 Cleaning Guidelines

CAUTION

Always power off the device and disconnect the adapter before cleaning.

Never use abrasive materials, bleach, hydrogen peroxide, or spray liquids directly onto the device.

- Chassis: Wipe the aluminium chassis with the included APEX Precision Cleaning Cloth lightly dampened with 70 percent isopropyl alcohol (IPA). The anodized aluminium finish is resistant to light solvents.
- Display: Use only a dry or very slightly damp microfiber cloth on the OLED panel. Never press hard on the panel surface, as OLED panels are susceptible to pressure damage. Avoid ammonia-based glass cleaners.
- Keyboard: Use compressed air (in short bursts, holding the can upright) to dislodge debris from between keys. Do not use vacuum cleaners directly on keys as this may dislodge keycaps.
- Ports: Use a dry, soft brush (such as a clean paintbrush) to remove dust from port openings. Never insert metal objects into ports.
- Vents: Use compressed air to blow out dust from the exhaust vents approximately every 3-6 months, depending on the environment.

15.2 Ventilation & Thermal Paste

- Ensure the bottom ventilation intakes are never obstructed during operation. Do not use the device on soft surfaces such as beds, sofas, or cushions that may block the vents.
- The liquid metal thermal interface material applied at the factory does not require replacement under normal usage conditions. APEX recommends inspection and re-application only after 3-5 years of heavy use, or if sustained thermal throttling is observed despite clean vents.
- Liquid metal re-application must only be performed by APEX-authorized service centers or highly experienced technicians. Liquid metal is electrically conductive and spillage onto the motherboard will cause permanent damage.

15.3 Storage Recommendations

- Short-term storage (1-30 days): Store the device at room temperature (15-25 degrees Celsius) in the included ProSleeve. Ensure the battery is at 40-60 percent charge.
- Long-term storage (30+ days): Charge the battery to approximately 50 percent and store in a cool, dry location away from direct sunlight. A storage temperature of 10-20 degrees Celsius is ideal.
- Do not store the device in an environment with rapid humidity fluctuations or near chemical solvents, as condensation or chemical vapour can cause corrosion of internal components.
- Never stack heavy objects on top of the device during storage.

CHAPTER 16

Troubleshooting

This chapter covers the most commonly encountered issues with the APEX PRO X1 Ultra and their recommended solutions. For issues not covered here, contact APEX Technical Support via the details in Chapter 18.

16.1 Boot Issues

Device does not power on

- Ensure the device is charged or connected to the adapter.
- Perform a hard reset: hold the power button for 15 seconds until the device powers off, then press it again.
- Disconnect all external peripherals (USB, Thunderbolt devices) and try again.
- If no response, the battery may be fully discharged. Connect the 240W adapter and wait 5 minutes before attempting to power on.

Device powers on but displays black screen

- Ensure the display brightness is not set to minimum (press Fn+F12 to increase brightness).
- Check if an external display is detected but the built-in display is disabled (press Fn+F10 to cycle display output modes).
- Perform a hard reset and attempt to boot into Safe Mode by pressing F8 repeatedly during boot.

Windows recovery screen appears on boot

- This indicates an abnormal previous shutdown. Select 'Continue to Windows 11' to attempt a normal boot.
- If the issue persists, use APEX DiagTools (bootable USB version, downloadable from apex-tech.com) to run memory and storage diagnostics.

16.2 Display Problems

Screen flickering	Disable HDR temporarily (Settings > System > Display > HDR). Update the GPU driver via APEX Update Manager. If using an external display, replace the cable.
Dead or stuck pixels	OLED pixel refresh can resolve stuck pixels: APEX Control Center > Display > OLED Care > Run Panel Refresh (takes ~5 minutes, device must remain connected to power).
Incorrect color output	Reset the color profile: APEX Control Center > Display > Color Profiles > Reset to Factory Calibration.
External display not detected	Ensure the cable and adapter are properly seated. Try a different port. Update GPU drivers. Cycle the external monitor power.

16.3 Network & Connectivity

- Wi-Fi not connecting: Ensure Wi-Fi is enabled (Action Center > Wi-Fi tile). Forget the network and reconnect. Restart the Wi-Fi adapter via Device Manager. Update the Intel Wi-Fi BE200 driver.
- Slow Wi-Fi performance: Ensure you are connecting to the 6 GHz band (SSID may differ). Enable MLO if your router supports Wi-Fi 7. Check for 2.4 GHz / 5 GHz interference.

- Bluetooth device not pairing: Remove the device from the paired devices list and re-initiate pairing. Ensure the Bluetooth device is in pairing mode. Toggle Bluetooth off and back on in Action Center.
- Thunderbolt device not recognized: Verify the Thunderbolt connection is secure. Update the Thunderbolt driver and approve the device in the Thunderbolt Security Centre (taskbar icon).

16.4 Performance Degradation

- Unexpected slowness: Check APEX Control Center for thermal throttling indicators. Ensure vents are clear of obstruction. Switch to APEX Performance or Turbo mode.
- High CPU temperature under light load: Close background applications, check for runaway processes in Task Manager, and run a malware scan.
- Fan running loudly at rest: This is normal immediately after waking from sleep as the system runs housekeeping tasks. If it persists for more than 5 minutes at idle, run APEX DiagTools.
- GPU not being utilized: Ensure the GPU mode is not set to iGPU Only in APEX Control Center. Verify that the application is configured to use the NVIDIA GPU in NVIDIA Control Panel > Manage 3D Settings.

CHAPTER 17

Technical Specifications

The following section provides the complete technical specification sheet for the APEX PRO X1 Ultra (Model: APX-X1U-2025). Specifications are subject to change without notice in line with component availability. Always verify current specifications at apex-tech.com/support/specifications.

17.1 Full Specification Sheet

PROCESSOR	
CPU Model	Intel Core Ultra 9 285HX
Cores / Threads	24 cores (8P + 16E + 4LP-E) / 32 threads
Base / Boost Clock	2.3 GHz base / 5.6 GHz max boost (P-Core)
Cache	36 MB Intel Smart Cache (L3)
Manufacturing Process	Intel 4 (7nm class EUV)
TDP (Configurable)	15W to 65W (base 45W)
NPU	Intel AI Boost — 48 TOPS
DISPLAY	
Panel	16-inch Samsung OLED, 3840x2400 (283 PPI)
Refresh Rate	Up to 120 Hz (adaptive)
Brightness (Peak HDR)	1,600 nits
Color Gamut	100% DCI-P3, 100% sRGB, 99% AdobeRGB
Delta E	< 1.0 (factory calibrated, X-Rite certified)
GRAPHICS	
Discrete GPU	NVIDIA GeForce RTX 5090 Ti Laptop, 24 GB GDDR7
Integrated GPU	Intel Arc 140V (32 Xe2 cores)
MEMORY	
Installed RAM	64 GB LPDDR5X-8533 (soldered, dual-channel)
Expansion	1x SODIMM DDR5-5600 (up to 32 GB, max system 96 GB)
STORAGE	
Primary SSD	4 TB Samsung 990 Pro PCIe 5.0 x4 NVMe 2.0
Expansion Slot	1x M.2 2280 PCIe 5.0 x4 (empty)
CONNECTIVITY	
Wi-Fi	Intel Wi-Fi 7 BE200 (802.11be, tri-band, 2x2 MIMO)
Bluetooth	5.4 (combo module with Wi-Fi)
PORTS	
Thunderbolt 5	2x (40 Gbps, DP 2.1, PD 140W)

USB-C 4.0	1x (DP 2.1 Alt Mode)
USB-A 3.2 Gen 2	2x (10 Gbps)
HDMI	1x HDMI 2.1 (10K @ 120Hz)
SD Card	1x UHS-III (985 MB/s)
Audio Jack	1x 3.5mm combo (Hi-Res, 384 kHz)
Charging	APEX MagCharge 240W (proprietary)
AUDIO	
Speakers	4x (2 tweeters + 2 woofers, 8W total, Harman Kardon)
Microphones	4x array with AI beamforming (APEX ClearVoice AI)
CAMERA	
RGB Camera	13 MP, f/2.0, 4K@30fps
IR Camera	Windows Hello face recognition
Privacy Shutter	Physical mechanical shutter with LED indicator
BATTERY & POWER	
Battery	99.9 Wh (8-cell Li-Po)
Adapter	240W GaN (included)
USB-C PD	Up to 140W charging
Battery Life	Up to 18 hours (video playback, Efficiency Mode)
SECURITY	
TPM	Infineon SLB 9672, TPM 2.0, fTPM
Fingerprint	Capacitive (power button integrated), FIDO2
Secure Boot	UEFI Secure Boot enabled by default
Lock Slot	Kensington Nano
OPERATING SYSTEM	
OS	Windows 11 Pro (64-bit), pre-installed and activated
PHYSICAL	
Dimensions	354.8 mm x 241.3 mm x 14.9 mm (W x D x H)
Weight	1.89 kg (without adapter)
Chassis Material	CNC-machined 6061-T6 aluminium alloy, Type III anodized
Color	Midnight Obsidian
MIL-STD	MIL-STD-810H certified (26 test methods)

CHAPTER 18

Warranty & Support

APEX Technologies provides a comprehensive warranty program for all APEX PRO X1 Ultra devices. The warranty reflects APEX's commitment to product quality and customer satisfaction across its global professional user base.

18.1 Limited Warranty Terms

Warranty Period	3 years (36 months) from the original date of purchase
Battery Warranty	2 years or 500 charge cycles (whichever comes first)
Accessories Warranty	1 year (power adapter, sleeve)
Coverage	Manufacturing defects in materials and workmanship
NOT Covered	Physical damage, liquid damage, unauthorized modifications, consumable parts (keycaps, rubber feet)
Onsite Service	Available in Tier 1 markets (Year 1: next-business-day onsite; Years 2-3: 5-business-day onsite)
Mail-in Service	Available globally. APEX provides a prepaid shipping label for warranty repairs.
Advanced Replacement	Available for APEX ProCare subscribers: replacement unit shipped before defective unit is returned.

WARRANTY ACTIVATION

Register your device within 30 days of purchase at apex-tech.com/register to activate the full 3-year warranty.

Proof of purchase (original invoice) must be retained for all warranty claims.

APEX Warranty covers India, United States, European Union, United Kingdom, Canada, Australia, Japan, and Singapore.

18.2 How to Contact Support

Live Chat & Ticket	apex-tech.com/support — available 24/7
Phone Support	+1-800-APEX-PRO (toll-free, US) +91-1800-APEX-IN (toll-free, India)
Email Support	pro-support@apex-tech.com
Response Time (Email)	Within 4 business hours for APEX PRO X1 Ultra owners
Community Forum	community.apex-tech.com
Support Hours (Phone)	Monday to Saturday, 09:00-21:00 local time

Service Centres (India)	Available in Mumbai, Delhi, Bangalore, Chennai, Hyderabad, Pune, and Ahmedabad
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18.3 International Service

- APEX maintains a network of authorized service centres in 42 countries. Use the Service Centre Locator at apex-tech.com/service to find the nearest facility.
- International warranty service is available for all APEX PRO X1 Ultra units regardless of the country of original purchase, provided the warranty has been registered.
- In markets without a physical service centre, mail-in service is provided via APEX's logistics partner with a fully insured collection and return process.
- Enterprise customers with APEX ProCare Enterprise contracts receive a dedicated account manager and 24/7 priority phone support with a 4-hour hardware response time SLA in Tier 1 markets.

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CHAPTER A

BIOS / UEFI Setup Reference

The APEX PRO X1 Ultra uses an AMI UEFI BIOS (version 3.x) with a custom APEX graphical interface. The BIOS provides low-level hardware configuration options that affect system boot behaviour, security, performance, and device management. Exercise caution when modifying BIOS settings; incorrect configuration can prevent the system from booting.

HOW TO ENTER BIOS SETUP

Power on or restart the device.

Press the Delete key or F2 key repeatedly during the APEX splash screen (within the first 2 seconds of POST).

Alternatively, in Windows: Settings > System > Recovery > Advanced Startup > Restart Now > Troubleshoot > Advanced Options > UEFI Firmware Settings > Restart.

A.1 Main Menu

System Date / Time	Set the real-time clock. Windows will sync this via NTP after boot.
BIOS Version	Displays current firmware version. Use APEX Update Manager to update; do not flash manually.
CPU Information	Read-only display of CPU model, core count, and microcode version.
Memory Information	Displays installed RAM capacity, speed, and slot configuration.
Storage Information	Lists detected NVMe and SATA devices with model and capacity.

A.2 Advanced Settings

Intel Virtualization Technology (VT-x)	Default: Enabled. Required for VMware, VirtualBox, Hyper-V, WSL 2.
Intel VT-d (IOMMU)	Default: Enabled. Required for PCIe device passthrough to virtual machines.
Hyper-Threading	Default: Enabled. Disabling may improve deterministic latency for audio production.
CPU Power Limits (PL1/PL2)	Default: Managed by APEX Control Center. Manual override available here.
SATA Controller Mode	Not applicable (no SATA devices present; M.2 slots are PCIe NVMe only).

USB Legacy Support	Default: Enabled. Allows USB keyboard/mouse during POST and in pre-OS environments.
Thunderbolt Security Level	Default: User Authorization. Options: No Security, User Authorization, Secure Connect, DisplayPort Only.
Intel Speed Shift Technology	Default: Enabled. Allows the OS to directly control CPU P-states for faster frequency scaling.
CFG Lock (MSR 0xE2)	Default: Locked. Advanced users running custom Linux kernels may need to unlock this.

A.3 Security Settings

Secure Boot	Default: Enabled (Standard Mode). Disable only for custom Linux deployments.
Secure Boot Mode	Default: Standard. Set to 'Custom' to enroll custom signing keys.
TPM Device	Default: Enabled (fTPM via Intel PTT). Discrete TPM also present.
TPM State	Default: Active. Can be Cleared (factory reset of TPM keys) from here.
Intel Platform Trust Technology (PTT)	Default: Enabled. PTT provides TPM 2.0 functionality via CPU firmware.
Administrator Password	Sets a BIOS access password. Required to enter BIOS setup if configured.
Hard Disk Password	Sets an ATA security password for the NVMe SSD (separate from BitLocker).
Boot Order Lock	Default: Disabled. Enabling prevents changes to boot device priority.

A.4 Power Settings

AC Behaviour	Default: Power On. Sets action when AC power is restored after outage.
AutoBoot on Lid Open	Default: Enabled. Device powers on automatically when lid is opened.
Wake on USB	Default: Disabled. Enables USB keyboard/mouse to wake device from Modern Standby.
Deep Sleep Control	Default: S0ix (Modern Standby). Legacy S3 Sleep available for compatibility.
Battery Charge Limit	Default: 100%. Set to 80% via APEX Control Center or directly here.
Fast Boot	Default: Enabled. Reduces POST time by skipping certain hardware checks.

A.5 Boot Menu

Boot Override	Temporarily boot from a selected device without changing the persistent boot order.
Boot Device Priority	Persistent boot order. Default: Windows Boot Manager, then APEX Recovery.
UEFI / Legacy Boot	Default: UEFI only. Legacy (CSM) mode is not supported on this platform.
Network Stack (PXE)	Default: Disabled. Enable for enterprise PXE network boot deployments.

BIOS RECOVERY

If the system becomes unbootable due to a failed BIOS flash, APEX BIOS Guard will automatically restore the last known-good BIOS version from the protected recovery partition.

For manual BIOS recovery: hold Fn + Power during cold boot to force recovery mode. Have a BIOS recovery USB drive ready (download from apex-tech.com/support/bios-recovery).

CHAPTER B

Glossary of Technical Terms

This glossary defines key technical terms used throughout this manual. Understanding these terms will help you make informed decisions when configuring and using your APEX PRO X1 Ultra.

AES-256

Advanced Encryption Standard with a 256-bit key length. A symmetric encryption algorithm adopted as a U.S. Federal standard, widely considered unbreakable with current computing capabilities.

Bandwidth (Memory)

The rate at which data can be read from or written to system memory, expressed in GB/s (Gigabytes per second). Higher bandwidth reduces bottlenecks in memory-intensive tasks such as video editing or large dataset processing.

Beamforming (Microphone)

A signal processing technique used in multi-microphone arrays to focus audio capture on the direction of the speaker, while simultaneously attenuating sound from other directions, effectively reducing background noise.

BitLocker

A full-volume encryption feature built into Microsoft Windows that uses AES-256 encryption to protect data on storage drives. It can use the TPM chip as a key protector to tie decryption to a trusted boot environment.

CUDA Cores

NVIDIA's general-purpose parallel computing processors within a GPU. A larger number of CUDA cores allows more parallel computations per clock cycle, improving performance in rendering, simulation, and AI inference tasks.

Delta E (dE)

A numerical measure of color accuracy. Delta E represents the difference between a displayed color and its target reference color. A value below 1.0 is imperceptible to the human eye and is the industry standard for professional-grade displays.

DCI-P3

A wide color gamut standard defined by the Digital Cinema Initiatives consortium, covering approximately 25% more colors than sRGB. It is the primary standard for digital cinema and is increasingly used for professional photography and video production.

DisplayPort 2.1 (DP 2.1)

The latest version of the DisplayPort standard, providing 80 Gbps of uncompressed bandwidth. It is capable of driving a single 16K display or multiple 8K displays simultaneously, and is available via Thunderbolt 5 and USB-C 4.0 on this device.

DLSS 4

NVIDIA Deep Learning Super Sampling version 4. An AI-powered upscaling and frame generation technology that uses Tensor Cores to generate high-quality high-resolution frames from lower-resolution inputs, significantly improving performance in supported applications.

fTPM (Firmware TPM)

A TPM 2.0 implementation that runs within the CPU's firmware rather than on a dedicated hardware chip. Intel Platform Trust Technology (PTT) is an fTPM implementation present in Intel processors.

GDDR7

The seventh generation of Graphics Double Data Rate memory, designed specifically for use as GPU VRAM. GDDR7 provides significantly higher bandwidth and energy efficiency compared to GDDR6X.

IOPS (Input/Output Operations Per Second)

A measure of storage device performance in random access scenarios. High IOPS is critical for operating system responsiveness, application launch times, and database operations.

Kensington Lock

A physical security standard for laptop locking slots and compatible security cables/locks. It allows the device to be physically secured to a desk or fixture using a steel cable lock.

LC3 Codec

Low Complexity Communications Codec. The audio codec specified for Bluetooth LE Audio (Bluetooth 5.2+). It provides improved audio quality and efficiency compared to the SBC codec used in classic Bluetooth audio.

LPDDR5X

Low Power Double Data Rate 5X memory. An evolution of LPDDR5 providing higher data rates (up to 8533 MT/s) with improved power efficiency, critical for extending battery life in thin-and-light laptops.

Liquid Metal TIM

Thermal Interface Material composed of a gallium-based alloy with exceptionally high thermal conductivity (typically 40-75 W/mK versus 4-8 W/mK for thermal paste). Applied between the CPU/GPU die and the heatsink/vapor chamber.

MIL-STD-810H

A testing standard developed by the United States Department of Defense to evaluate equipment durability under extreme environmental conditions. 26 test methods include thermal shock, altitude, vibration, shock, and humidity.

MLO (Multi-Link Operation)

A Wi-Fi 7 feature that enables simultaneous data transmission and reception across multiple frequency bands (e.g., 5 GHz and 6 GHz simultaneously), reducing latency and improving reliability.

NVMe (Non-Volatile Memory Express)

A host controller interface and storage protocol designed specifically for SSDs communicating over the PCIe bus. NVMe provides lower latency and higher throughput compared to the AHCI protocol used by legacy SATA SSDs.

OLED (Organic LED)

A display technology where each pixel consists of organic compounds that emit light when an electric current is applied. Unlike LCD displays, OLED pixels can switch off individually, producing true black and an infinite contrast ratio.

PCIe 5.0 (PCI Express 5.0)

The fifth generation of the PCIe interconnect standard, providing 128 GB/s bandwidth per x16 slot, double that of PCIe 4.0. PCIe 5.0 x4 NVMe SSDs achieve sequential read speeds up to 14,500 MB/s.

Secure Boot

A UEFI firmware security feature that ensures only software cryptographically signed by trusted authorities can execute during the boot process, protecting against bootloader-level malware and rootkits.

SODIMM

Small Outline Dual Inline Memory Module. A compact form factor RAM module used in laptops and small form factor computers. The APEX PRO X1 Ultra has one SODIMM expansion slot accepting DDR5 modules.

TBW (Terabytes Written)

A measure of SSD endurance rating — the total amount of data that can be written to the drive before NAND flash wear is expected to cause failures. The Samsung 990 Pro 4 TB is rated at 3,600 TBW.

Tensor Cores

Specialized processing units within NVIDIA GPUs designed to accelerate matrix multiplication operations, which are fundamental to AI and machine learning workloads. 5th generation Tensor Cores in the RTX 5090 Ti support FP8, FP16, BF16, and INT8 precision formats.

Thunderbolt 5

Intel's latest Thunderbolt standard providing 40 Gbps bidirectional bandwidth (scalable to 120 Gbps TX in Bandwidth Boost mode), DisplayPort 2.1 Alt Mode, USB4, and Power Delivery up to 140W in a single USB-C connector.

TIM (Thermal Interface Material)

Any substance placed between a heat-generating component (CPU/GPU die) and its heatsink to eliminate microscopic air gaps and improve heat transfer. Common types include thermal paste and liquid metal.

TPM 2.0 (Trusted Platform Module)

An international standard (ISO/IEC 11889) defining a dedicated hardware chip or firmware module for cryptographic operations. TPM 2.0 provides secure key generation, storage, attestation, and random number generation, forming the root of trust for BitLocker and Windows Hello.

UHS-III (Ultra High Speed Phase III)

The fastest SD card bus interface specification, providing up to 624 MB/s full-duplex speed. Required to achieve maximum read/write speeds from V90-rated SD cards used in professional video cameras.

Vapor Chamber

An advanced heat spreader technology consisting of a sealed, flat chamber containing a small amount of working fluid. Heat causes the fluid to vaporize at the hot spot, travel through the chamber, condense at cooler regions, and return via a wick structure, spreading heat very efficiently over a large area.

VRR (Variable Refresh Rate)

A display technology that dynamically synchronizes the display's refresh rate to the GPU's frame output rate, eliminating screen tearing without introducing the input lag associated with traditional V-Sync.

WPA3

Wi-Fi Protected Access 3. The latest generation of the Wi-Fi Alliance's wireless security certification, providing stronger encryption (192-bit security suite for WPA3-Enterprise) and protection against offline dictionary attacks via Simultaneous Authentication of Equals (SAE).

CHAPTER C

Complete Keyboard Shortcut Reference

This appendix provides a complete reference for all keyboard shortcuts available on the APEX PRO X1 Ultra, organized by category. These shortcuts are in addition to standard Windows 11 keyboard shortcuts and are specific to the APEX hardware.

C.1 APEX Hardware Shortcuts (Fn Key Combinations)

Fn + F1	Mute / unmute audio output
Fn + F2	Decrease speaker volume
Fn + F3	Increase speaker volume
Fn + F4	Hardware microphone mute/unmute (amber LED on key)
Fn + F5	Refresh (browser) / Recalculate (spreadsheet)
Fn + F6	Toggle physical camera privacy shutter
Fn + F7	Previous media track
Fn + F8	Play / Pause media
Fn + F9	Next media track
Fn + F10	Cycle display output modes
Fn + F11	Decrease display brightness
Fn + F12	Increase display brightness
Fn + Spacebar	Cycle keyboard backlight: Off > Low > Medium > High > RGB
Fn + Q	Cycle APEX Performance Mode: Efficiency > Balanced > Performance > Turbo
Fn + Del	Screenshot (saved to clipboard and Screenshots folder)
Fn + Z	Toggle APEX Therma-Boost AI between Auto and Manual
Fn + 1 / 2 / 3	Apply Keyboard RGB Preset Profiles 1, 2, or 3
Fn + Lock	Toggle Fn key lock (Fn LED illuminates when locked)
Fn + Esc	Open APEX Control Center
Fn + Up/Down Arrow	Increase / decrease keyboard backlight brightness
Fn + Left/Right Arrow	Decrease / increase display brightness (alternate)

C.2 Essential Windows 11 Shortcuts

Win + A	Open Quick Settings (Action Center)
Win + D	Show / hide Desktop
Win + E	Open File Explorer
Win + I	Open Windows Settings

Win + K	Open Cast / Connect sidebar
Win + L	Lock the screen
Win + N	Open Notification Centre
Win + P	Project display options (Duplicate, Extend, Second Screen Only)
Win + R	Open Run dialog
Win + S	Open Windows Search
Win + V	Open Clipboard History
Win + W	Open Widgets panel
Win + X	Open Quick Link (Power User) menu
Win + Z	Open Snap Layouts grid for active window
Win + Tab	Open Task View (all virtual desktops and windows)
Win + Ctrl + D	Create a new Virtual Desktop
Win + Ctrl + Left/Right	Switch between Virtual Desktops
Win + Ctrl + F4	Close current Virtual Desktop
Win + Shift + S	Open Snipping Tool for partial screenshot
Alt + Tab	Switch between open application windows
Alt + F4	Close the active application window
Ctrl + Shift + Esc	Open Task Manager directly
Ctrl + Alt + Del	Open Windows Security screen

CHAPTER D

Environmental Specifications & Compliance

This appendix details the environmental operating conditions, storage conditions, and compliance certifications applicable to the APEX PRO X1 Ultra. These specifications are important for deployment planning in environments with specific climate or regulatory requirements.

D.1 Operating & Storage Conditions

Operating Temperature	5 degrees C to 40 degrees C (41 degrees F to 104 degrees F)
Storage Temperature	-20 degrees C to 60 degrees C (-4 degrees F to 140 degrees F)
Operating Humidity	10% to 90% RH (non-condensing)
Storage Humidity	5% to 95% RH (non-condensing)
Operating Altitude	0 to 5,000 metres (0 to 16,404 feet) above sea level
Storage Altitude	0 to 15,240 metres (0 to 50,000 feet)
Vibration (Operating)	0.66 GRMS (5-500 Hz, random profile)
Vibration (Non-Operating)	1.30 GRMS (5-500 Hz, random profile)
Shock (Operating)	40G, 11ms, half-sine waveform
Shock (Non-Operating)	200G, 2ms, half-sine waveform
Ingress Protection	IPX3 rated (keyboard spill resistance, 60mL, 15 seconds)

D.2 Regulatory & Safety Certifications

FCC Part 15 Class B	United States — Electromagnetic interference limits for consumer devices.
IC (Industry Canada) Class B	Canada — Equivalent to FCC Class B for the Canadian market.
CE Mark (Europe)	European Union — Compliance with RED 2014/53/EU, LVD 2014/35/EU, EMC 2014/30/EU, RoHS 2011/65/EU.
UKCA Mark	United Kingdom — Post-Brexit conformity marking equivalent to CE for the UK market.
BIS (Bureau of Indian Standards)	India — Mandatory certification under CRS (Compulsory Registration Scheme) for electronics.
MIC (Ministry of Internal Affairs)	Japan — Radio equipment certification for Wi-Fi and Bluetooth operations.
KC Mark (Korea)	Republic of Korea — Electromagnetic compliance and safety certification.

RCM (Regulatory Compliance Mark)	Australia and New Zealand — Electrical safety and EMC compliance.
BSMI (Taiwan)	Taiwan — Bureau of Standards, Metrology and Inspection product safety certification.
CCC (China Compulsory Certification)	People's Republic of China — Mandatory safety and EMC certification.
ENERGY STAR 8.0	International — Energy efficiency certification for laptop computers.
EPEAT Gold	International — Environmental performance criteria for electronic products.
TCO Certified Edge	International — Sustainability and ergonomic performance certification.
TUV Rheinland Low Blue Light	International — Certification for display blue light emission reduction.
RoHS (Restriction of Hazardous Substances)	International — Restriction of lead, mercury, cadmium, hexavalent chromium, PBB, PBDE.
REACH Compliance	European Union — Regulation on chemicals and substances in materials.
Conflict Mineral Free (3TG)	International — Tantalum, tin, tungsten, and gold sourced from conflict-free smelters.

D.3 Environmental Commitment

APEX Technologies is committed to reducing the environmental footprint of its products throughout their entire lifecycle, from raw material sourcing through end-of-life recycling.

- **Packaging:** 100% of outer packaging is made from FSC-certified recycled cardboard. All foam inserts have been replaced with moulded pulp fibre inserts. The Pro Sleeve is manufactured from recycled polyester (rPET) fabric.
- **Chassis Recycled Content:** The aluminium alloy chassis contains a minimum of 40% post-industrial recycled aluminium, reducing the energy consumption of raw material production by approximately 95% compared to primary aluminium smelting.
- **PVC and Brominated Flame Retardant Free:** All cables, connectors, and printed circuit boards in the APEX PRO X1 Ultra are free from polyvinyl chloride (PVC) and brominated flame retardants (BFR).
- **ENERGY STAR Compliance:** The APEX PRO X1 Ultra meets ENERGY STAR 8.0 requirements, consuming up to 30% less energy in typical use scenarios compared to a standard non-certified laptop.
- **End-of-Life Recycling:** APEX participates in take-back and recycling programmes in all major markets. Visit apex-tech.com/recycling to find a drop-off point or request a prepaid recycling mailer for your old APEX device.
- **Repairability:** The APEX PRO X1 Ultra scores 7.5 out of 10 on the iFixit Repairability Scale, reflecting accessible upgrade paths (RAM, SSD) and publicly available service manuals on the APEX Support portal.

D.4 Acoustic Emission

Sound Pressure Level (Idle)	0 dB(A) (fans off at idle in Efficiency Mode)
Sound Pressure Level (Balanced Load)	28 dB(A) at 1 metre
Sound Pressure Level (Maximum Fan RPM)	43 dB(A) at 1 metre

Measurement Standard	ISO 7779:2018 (Acoustics — Measurement of airborne noise from IT equipment)
Measurement Position	Operator position, 1 metre bystander position

End of Document | APEX PRO X1 Ultra Official User Manual | Revision 1.0 | April 2026 | Document No. APX-UM-2025-001 For the latest version of this manual, firmware downloads, and support resources, visit apex-tech.com/support. This document is also available in French, German, Spanish, Japanese, Korean, Simplified Chinese, Traditional Chinese, and Hindi at apex-tech.com/manuals.

CHAPTER C

Complete Keyboard Shortcut Reference

This appendix provides a complete reference for all keyboard shortcuts available on the APEX PRO X1 Ultra, organized by category. These shortcuts are in addition to standard Windows 11 keyboard shortcuts and are specific to the APEX hardware.

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Fn + F7	Previous media track
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Fn + F10	Cycle display output modes
Fn + F11	Decrease display brightness
Fn + F12	Increase display brightness
Fn + Spacebar	Cycle keyboard backlight: Off > Low > Medium > High > RGB
Fn + Q	Cycle APEX Performance Mode: Efficiency > Balanced > Performance > Turbo
Fn + Del	Screenshot (saved to clipboard and Screenshots folder)
Fn + Z	Toggle APEX Therma-Boost AI between Auto and Manual
Fn + 1	Apply Keyboard Preset Profile 1
Fn + 2	Apply Keyboard Preset Profile 2
Fn + 3	Apply Keyboard Preset Profile 3
Fn + Lock	Toggle Fn key lock (Fn LED on Fn key illuminates when locked)
Fn + Esc	Open APEX Control Center

Fn + Up/Down Arrow	Increase / decrease keyboard backlight brightness
Fn + Left/Right Arrow	Decrease / increase display brightness (alternate)

C.2 Essential Windows 11 Shortcuts

Win + A	Open Quick Settings (Action Center)
Win + D	Show / hide Desktop
Win + E	Open File Explorer
Win + G	Open Xbox Game Bar (performance overlay)
Win + H	Open Voice Typing
Win + I	Open Windows Settings
Win + K	Open Cast / Connect sidebar
Win + L	Lock the screen
Win + M	Minimize all windows
Win + N	Open Notification Centre
Win + P	Project display options (Duplicate, Extend, Second Screen Only)
Win + R	Open Run dialog
Win + S	Open Windows Search
Win + U	Open Accessibility settings
Win + V	Open Clipboard History
Win + W	Open Widgets panel
Win + X	Open Quick Link menu (Power User menu)
Win + Z	Open Snap Layouts grid for the active window
Win + . (period)	Open Emoji picker
Win + Spacebar	Switch input language / keyboard layout
Win + Tab	Open Task View (all virtual desktops and windows)
Win + Ctrl + D	Create a new Virtual Desktop
Win + Ctrl + Left/Right	Switch between Virtual Desktops
Win + Ctrl + F4	Close current Virtual Desktop
Win + Shift + S	Open Snipping Tool for partial screenshot
Win + PrtScn	Full screenshot saved to Pictures > Screenshots folder
Alt + Tab	Switch between open application windows
Alt + F4	Close the active application window
Ctrl + Shift + Esc	Open Task Manager directly
Ctrl + Alt + Del	Open Windows Security screen

C.3 Productivity & Browser Shortcuts

Ctrl + Z / Y	Undo / Redo — universal across most applications
Ctrl + A	Select All
Ctrl + C / X / V	Copy / Cut / Paste
Ctrl + F	Find text (browser, document, file explorer)
Ctrl + S	Save the current document or file
Ctrl + P	Print (or open Print Preview dialog)
Ctrl + W	Close the active tab or window
Ctrl + T	Open a new tab (browsers and some applications)
Ctrl + Tab	Cycle through open tabs (right)
Ctrl + Shift + Tab	Cycle through open tabs (left)
Ctrl + N	New window / new document
Ctrl + Shift + N	New incognito / private window (Chrome, Edge, Firefox)
Ctrl + L	Focus the address bar (browsers)
Ctrl + D	Bookmark the current page (browsers)
Ctrl + H	Open browsing history (browsers)
Ctrl + J	Open downloads panel (browsers)
Ctrl + Shift + I	Open Developer Tools (browsers)
F5 / Ctrl + R	Refresh the current page or view
Ctrl + Shift + R	Hard refresh (bypass cache) in browsers
Alt + Left Arrow	Navigate back (browsers and File Explorer)
Alt + Right Arrow	Navigate forward (browsers and File Explorer)
F11	Toggle full-screen mode (browsers)
F12	Open Developer Tools / Network Inspector (browsers)